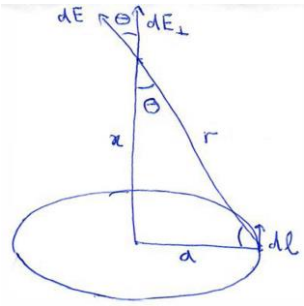
Exam Solutions 2016 – PST

[(3 + 2) + (5 + 3 + 2) = 15 marks]

Q1.

(a) (i)



$dE = \frac{dq}{4\pi\varepsilon_0 r^2}$, $r = \sqrt{a^2 + x^2}$

Horizontal component of field cancels leaving

$dE_{\text{vert}} = dE cos\theta = dE\frac{x}{r} = \frac{xdq}{4\pi\varepsilon_0 r^3} = \frac{xdq}{4\pi\varepsilon_0 (a^2+x^2)^{\frac{3}{2}}}$.

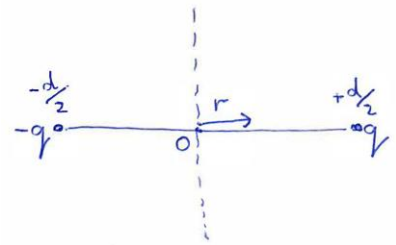
$Q = \lambda 2\pi a$ => $dq = \lambda dl$ => $dE_{\text{vert}} = \frac{\lambda x dl}{4\pi\varepsilon_0 (a^2+x^2)^{\frac{3}{2}}}$

$E_{\text{vert}} = \frac{\lambda x}{4\pi\varepsilon_0 (a^2+x^2)^{\frac{3}{2}}} \int_0^{2\pi a} dl = \frac{2\pi a\lambda x}{4\pi\varepsilon_0 (a^2+x^2)^{\frac{3}{2}}} = \frac{Qx}{4\pi\varepsilon_0 (a^2+x^2)^{\frac{3}{2}}}$

(ii) In the limit x>>a

$E = \frac{Qx}{4\pi\varepsilon_0 (a^2+x^2)^{\frac{3}{2}}} \rightarrow \frac{Qx}{4\pi\varepsilon_0 (x^2)^{\frac{3}{2}}} = \frac{Q}{4\pi\varepsilon_0 x^2}$, i.e. point charge at distance x.

(b) (i) **[1 mark diagram]**



$\vec{E} = \frac{-q}{4\pi\varepsilon_0 \left(\frac{d}{2}+r\right)^2}(-\hat{\hat{r}}) + \frac{+q}{4\pi\varepsilon_0 \left(\frac{d}{2}-r\right)^2}(-\hat{\hat{r}})$

$= \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{\left(\frac{d}{2}+r\right)^2} - \frac{1}{\left(\frac{d}{2}-r\right)^2}\right)\hat{\hat{r}} = \frac{q}{4\pi\varepsilon_0}\left(\frac{2dr}{\left(\frac{d}{2}+r\right)^2\left(\frac{d}{2}-r\right)^2}\right)\hat{\hat{r}}$

$= \frac{qd}{2\pi\varepsilon_0}\left(\frac{r}{\left(\frac{d^2}{4}-r^2\right)^2}\right)\hat{\hat{r}}$, $\mu = qd$

⇨ $E = \frac{\mu}{2\pi\varepsilon_0}\left(\frac{r}{\left(\frac{d^2}{4}-r^2\right)^2}\right)$, r>>d => $\left(\frac{d^2}{4} - r^2\right) \approx r^2$

⇨ $E \approx \frac{\mu}{2\pi\varepsilon_0 r^3}$

(ii) **[field from +ve -]**

[field to -ve -]

[general shape correct -]

(iii) $U = -\vec{\mu}.\vec{E}$

$= \frac{\mu^2}{2\pi\varepsilon_0 r^3}$

Q2

i) Use Gauss's law, such that Flux=0 outside the capacitor and =E.A inside the capacitor.
=> $E=q/A\varepsilon_o$ $V=qd/A\varepsilon_o$, so $C=A\varepsilon_o/d$

ii) $A=Cd/\varepsilon_o = 1.13 \times 10^5\ m^2$

iii) Using Gauss's Law – $E=q/2\pi r l\varepsilon_o$, $V=q/2\pi l\varepsilon_o \ln(b/a)$, $C=2\pi l\varepsilon_o/\ln(b/a)$

iv) $\varepsilon=\varepsilon_o\kappa$ => $C=2\pi l\kappa\varepsilon_o/\ln(b/a)$

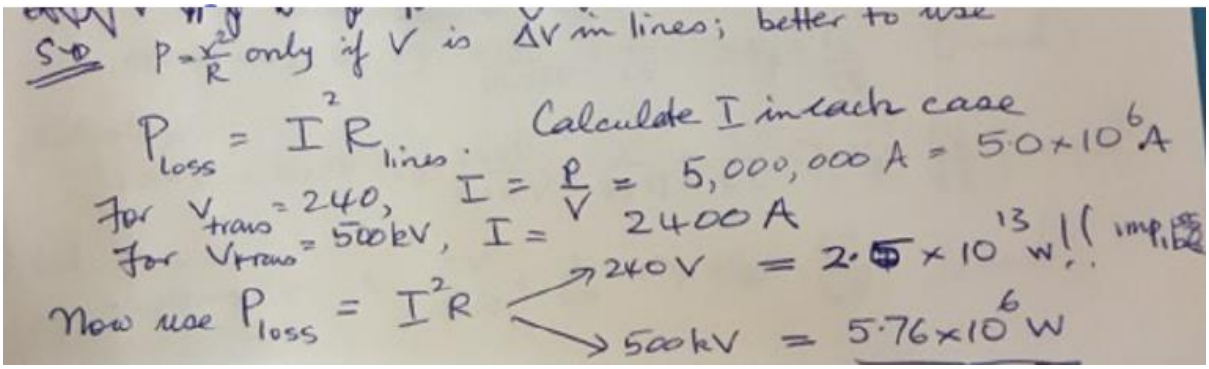
Q3

a) Central current = 0, => current through R1 and R2 same, current through Rv and R3 also equal.

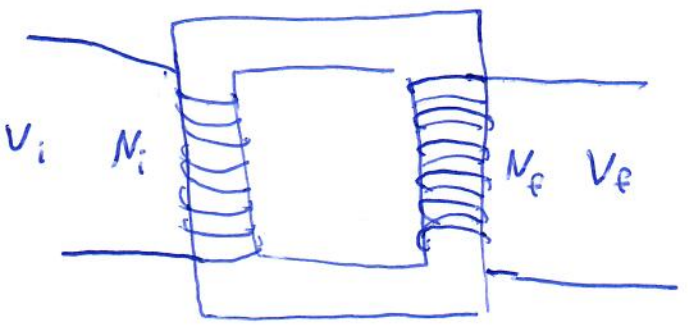
Use Kirchoff loops - $-i_1R_1 + i_2R_v = 0$, $-i_1R_2 + i_2R_3 = 0$

⇨ $R_v = R_3 (R_1/R_2)$

b) I)



(ii) Transformer



modelled as N loops of wire in series, each with potential difference $\frac{V}{N}$ accross it.

A loop of wire with ΔV around it leads to changing magnetic flux by Faraday's Law $\left(\mathcal{E} = -\frac{d\Phi_B}{dt}\right) \Rightarrow \frac{d\Phi_B}{dt} = \frac{V}{N}$ per loop.

$\Rightarrow$ Total ~~the~~ rate of change of magnetic flux on left hand side is : $N_i \frac{d\Phi_{Bi}}{dt} = V_i$ [1 mark]

$\frac{d\Phi_B}{dt} = \frac{dB}{dt} A$, where A is the area of an individual loop

A is the same on the left hand side to the right hand side of ~~the~~ transformer, as is $\frac{dB}{dt} \Rightarrow \frac{d\Phi_{Bi}}{dt} = \frac{d\Phi_{Bf}}{dt} = \frac{d\Phi_B}{dt}$ [1 mark]

$\Rightarrow \frac{d\Phi_B}{dt} = \frac{V_i}{N_i} = \frac{V_f}{N_f} \Rightarrow \frac{N_f}{N_i} = \frac{V_f}{V_i}$ [1 mark]

1st change: ~~500kV~~

$500kV \rightarrow 20kV \Rightarrow \frac{20kV}{500kV} = \frac{1}{25} \Rightarrow N_f = \frac{N_i}{25}$ (1 mark)

2nd change:

$20kV \rightarrow 240V \Rightarrow \frac{240V}{20 \times 10^3 V} = \frac{1}{83.33} \approx \frac{1}{83} \Rightarrow N_f \approx \frac{N_i}{83}$ [1 mark]

3rd change

$240V \rightarrow 5V \Rightarrow \frac{5V}{240V} = \frac{1}{48} \Rightarrow N_f = \frac{N_i}{48}$ [1 mark]

Q4.

[(3+3+3) + (3+3) = 15 marks]

(a) (i) Lorentz force $\underset{\sim}{F} = q\underset{\sim}{v} \times \underset{\sim}{B} = qvB\,\hat{\underset{\sim}{F}}$

where $\hat{\underset{\sim}{F}}$ is $\perp$ to $\hat{\underset{\sim}{v}}$ and $\hat{\underset{\sim}{B}}$.

$\hat{\underset{\sim}{F}} \perp$ to $\hat{\underset{\sim}{v}}$ $\Rightarrow$ centripetal force.

$\Rightarrow F = qvB = \dfrac{mv^2}{r}$ [1 mark]

$\Rightarrow \dfrac{v}{r} = \dfrac{qB}{m}$, Now $\dfrac{v}{r} = \omega$ ← angular frequency [1 mark]
$= 2\pi f$

$\Rightarrow 2\pi f = \dfrac{qB}{m} \Rightarrow f = \dfrac{qB}{2\pi m}$ [1 mark]

(ii) • B does no work $\Rightarrow$ E is required to increase $|\underset{\sim}{v}|$

• $\underset{\sim}{F} = q\underset{\sim}{E} = m\underset{\sim}{a} = m\dfrac{d\underset{\sim}{v}}{dt} \Rightarrow \underset{\sim}{E}$ must be in same direction as $\underset{\sim}{v}$.

• Applying E at the same point in the charges'

~~orbit will accelerate the charge whose direction of E must be applied at~~

orbit will give the charge a "kick" each time around, increasing $|\underset{\sim}{v}|$, i.e. the speed in the direction of travel.

• E must therefore oscillate with the orbital frequency.

[3 marks - subset of these or equivalent arguments.]

(iii) $K.E. = \frac{1}{2}mv^2$, K.E. max when v max. [½ mark]

Circular motion $\Rightarrow$ $v = \frac{qBr}{m}$ [1 mark]

q, B, m all constants.

$\Rightarrow$ v_{max} when r_{max} [½ mark]

$\Rightarrow$ $v_{max} = \frac{qBR}{m}$ $\Rightarrow$ $K.E._{max} = \frac{q^2B^2R^2}{2m}$ [1 mark]

(b)(i)

[1 mark]

$d\underset{\sim}{B} = \frac{\mu_0 i}{4\pi}\frac{d\underset{\sim}{s}\times\hat{\underset{\sim}{r}}}{r^2}$ $\Rightarrow$ $d\underset{\sim}{B}_\perp = \frac{\mu_0 i}{4\pi}\frac{d\underset{\sim}{s}\times\hat{\underset{\sim}{r}}}{r^2}\underbrace{\cos\theta}_{\frac{R}{r}}$

$d\underset{\sim}{B}$ ∥ cancel $\Rightarrow$ $\hat{\underset{\sim}{B}} = \hat{\underset{\sim}{B}}_\perp = \hat{\underset{\sim}{x}}$

$\Rightarrow$ $\underset{\sim}{B} = \int_0^{2\pi R}\frac{\mu_0 i}{4\pi}\frac{R}{r^3}\,ds$

$= \frac{\mu_0 i}{2\cancel{4\pi}}\frac{\cancel{2\pi}R^2}{r^3}\hat{\underset{\sim}{x}}$, $r^2 = R^2+x^2$

[1 mark – work]

$\Rightarrow$ $B = \frac{\mu_0 R^2}{2(R^2+x^2)^{3/2}}$ [1 mark]

be generous! Q actually (mistakenly) asks to use Ampere's law!

(ii) $\mu = iA$ [1 mark]

$= ef(\pi R^2)$ [1 mark]

$= 1.6 \times 10^{-19}\ C \times 6.5 \times 10^{15}\ Hz \times \pi \times (5.3 \times 10^{-11}\ m)^2$

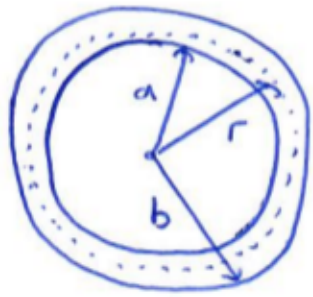
$= 9.2 \times 10^{-24}\ A \cdot m^2$ [1 mark]

Q5.

[(1+3+1) + (5+5) = 15 marks]

(a) (i) $B \approx 0$ [1 mark]

(ii)



Ampere's Law:

$\oint \underset{\sim}{B} \cdot d\underset{\sim}{s} = \mu_0 i_{enc}$

Amperian loop of radius r,

$i_{enc} = Ni$, N = no. of turns. [½ mark]

$\oint \underset{\sim}{B} \cdot d\underset{\sim}{s} = B \oint ds = 2\pi r B$ [½ mark]

$\Rightarrow B 2\pi r = \mu_0 N i$

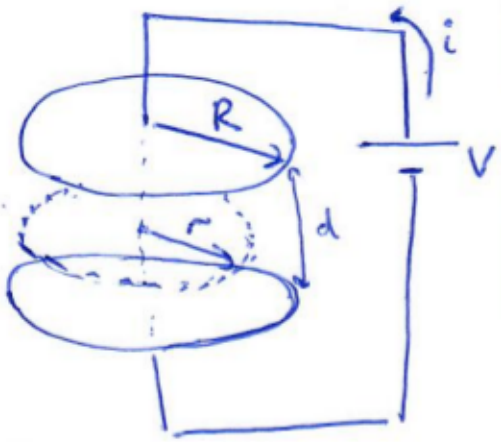
$\Rightarrow B = \dfrac{\mu_0 N i}{2\pi r}$ [1 mark]

direction: CW [1 mark]

(iii) $B \approx 0$ [1 mark]

(b) (i)

[1 mark]



Ampere's Law: $\oint \underset{\sim}{B} \cdot d\underset{\sim}{s} = \mu_0 i_{enc} + \mu_0 \varepsilon_0 \dfrac{d\Phi_E}{dt}$

Using Amperian loop, radius $r < R \Rightarrow \oint \underset{\sim}{B} \cdot d\underset{\sim}{s} = 2\pi r B$ } [1 mark]

$i_{enc} = 0$, $\Phi_E = \underset{\sim}{E} \cdot \underset{\sim}{A} = E\pi r^2$ ($r < R \Rightarrow E \neq 0$)

$\Rightarrow 2\pi r B = \dfrac{d(E\pi r^2)}{dt} \mu_0 \varepsilon_0 \Rightarrow B = \dfrac{\mu_0 \varepsilon_0 r}{2} \dfrac{dE}{dt}$ [1 mark]

(ii) Ampere's Law: $\oint \underset{\sim}{B} \cdot d\underset{\sim}{s} = \mu_0 i_{enc} + \mu_0 \varepsilon_0 \frac{d\phi_E}{dt}$

Amperian loop, as before but $r > R$. $\Rightarrow \oint \underset{\sim}{B} \cdot d\underset{\sim}{s} = 2\pi r B$

$i_{enc} = 0$, $\phi_E = \underset{\approx}{E} \cdot \underset{\sim}{A}$, now $E \begin{cases} = 0, & r > R \\ \neq 0, & r \leq R \end{cases}$ [1 mark]

$\Rightarrow \phi_E = \phi_E\big|_{r=R} + \phi_E\big|_{r>R} = E\pi R^2 + 0$ [1 mark]

$\Rightarrow \frac{d\phi_E}{dt} = \frac{dE}{dt}\pi R^2 \Rightarrow 2\pi r B = \frac{\mu_0 \varepsilon_0 R^2}{2r}\frac{dE}{dt}$ [2 mark]

[3 + 4 + 4 = 11 marks]

Q6.

(i) $F = PA = P_0A + \rho ghA$

$F = 13.74\ N$

(ii) $P + \frac{1}{2}\rho v^2 + \rho gh =$ P +

when water exits hole

⇨ $\Delta P = \frac{1}{2}\rho\Delta(v^2),\ h = \text{const}$

$\Delta P = \rho g h_{\text{hole}}$

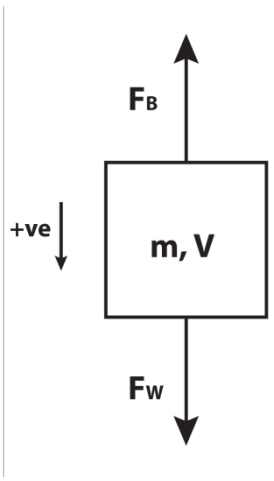
⇨ $v_f = \sqrt{2gh_{\text{hole}}}$

⇨ $= \sqrt{2 * 9.8\text{ms}^{-2} * 2\text{m}} = 6.26\text{ms}^{-1}$

volume = 10^4 m^3; flow rate in m^3/s = Av = $6.26 \times \pi \left(\frac{1.2*10^{-2}}{2}\right)^2$ =0.000708

hence t = 1.41×10^7 s

(iii) **[1 mark diagram]**



$F_{net} = F_W - F_B = mg - \rho_w gV$

$= (m - \rho V)g = (500\text{kg} - 1030\text{kgm}^{-3} * 0.5\text{m}^3) * 9.8\text{ms}^{-1}$

$= -147$ N

net force is upwards, i.e. this will float in seawater; no force is required

[5 + (2 + 4) + (2 + 1) = 14 marks]

Q7.

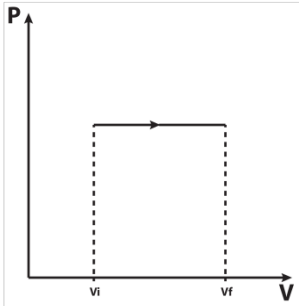
(a) $Q = Q_L + Q_T^{water} + Q_T^{ice}$

$= mL + c_w m\Delta T_w + c_I m\Delta T_I$

=396.1 kJ

(b) (i

(c) (ii)



(ii) Does work

work ON gas given by $W = -\int_{V_i}^{V_f} PdV$, P=const => $W = -P\Delta V$,

$\Delta V > 0$ => $W < 0$

$=> Work\ done\ BY\ gas$

(d) (i) Heat in = ΔU

= 300 J

(ii) $\Delta V = 0, => W = 0$

[(3 + 4) + 4 + (2 + 4 + 3) = 20 marks]

Q8.

(a) (i) $hf = \frac{hc}{\lambda} = W + K.E.$, W is the work function.

will exhibit photoelectric effect if $hf = \frac{hc}{\lambda} > W$

$\frac{hc}{\lambda} = \frac{4.14*10^{-15}\text{eV.s}*3*10^{8}\text{ms}^{-1}}{400*10^{-9}\text{m}} = 3.105\ \text{eV}$

Thus only Lithium will exhibit photoelectric effect.

(ii) $K.E. = \frac{hc}{\lambda} - W$

$= 0.81\ \text{eV}$

the others will be zero.

(b) $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}}$

$2dsin\theta = m\lambda$ => $d = \frac{m\lambda}{2sin\theta} = \frac{mh}{2\sqrt{2mK}sin\theta}$

$= 1.09 * 10^{-10}\ \text{m}$

(c) (i) $\Delta x \Delta p \geq \frac{h}{4\pi}$

$\Delta p = 10^{-6}\ \text{kg ms}^{-1}$ => $\Delta x * 10^{-6}\ \text{kg ms}^{-1} \geq \frac{h}{4\pi}$

$\Delta x \geq 5.28 * 10^{-29}\ \text{m}$

(ii) $\Delta v = \frac{\Delta p}{m} = 2.5 * 10^{-7}\text{ms}^{-1}$

⇨ $\Delta p_e = m_e \Delta v$

$= 2.28 * 10^{-37}\text{ kg ms}^{-1}$

⇨ $\Delta x_e \geq \frac{h}{4\pi\Delta p_e}$

⇨ $\Delta x_e \geq 231.4\text{ m}$

(iv) - ball has insignificant x uncertainty, while electron has substantial x uncertainty

-macroscopic objects have negligible position uncertainty because of their large masses

-microscopic objects have large position uncertainty because of their small masses

[(3 + 3) + (3 + 2 + 2 + 3) = 16 marks]

Q9.

(a) **(i)** $f = \frac{c}{\lambda} = cR(\frac{1}{n_2^2} - \frac{1}{n_1^2})$

⇨ $\frac{1}{n_1^2} = \frac{1}{n_2^2} - \frac{1}{R\lambda}$

Lyman series, i.e. n_2=1

$n_1 = \sqrt{\frac{R\lambda}{R\lambda - 1}}$

$= 2$

(ii) $\frac{1}{\lambda} = R(\frac{1}{n_2^2} - \frac{1}{n_1^2})$

⇨ λ is minimum when $(\frac{1}{n_2^2} - \frac{1}{n_1^2})$ is maximum

Balmer series so n_2=2 => $\frac{1}{\lambda} = R(\frac{1}{4} - \frac{1}{n_1^2})$.

clearly $(\frac{1}{4} - \frac{1}{n_1^2})$ is maximum when $\frac{1}{n_1^2} = 0$, which only happens as $n_1 \to \infty$

⇨ $n_1 \to \infty$ leads to λ_{min}

$\lambda_{min} = \frac{4}{R} = 365\text{nm}$

(b) (i) Circumference of circle = integer no. of wavelengths

⇨ $2\pi r = n\lambda$

De-Broglie wavelength $\lambda = \frac{h}{p} = \frac{h}{mv}$

⇨ $2\pi r = \frac{nh}{mv}$ => $mvr = \frac{nh}{2\pi}$

(ii) $r_n = \frac{\varepsilon_0 h^2}{\pi m e^2} n^2$, n=6

$= 1.9\text{ nm}$

alternatively

multiply Bohr radius (52.92 pm on formula sheet) by 36 = 1.9 nm

(iii) $v_n = \frac{nh}{2\pi m r_n}$

$= 3.66 * 10^5 \mathrm{ms}^{-1}$

alternatively

use $v_n = c\alpha_{\mathrm{em}} \frac{1}{n}$, $\alpha_{\mathrm{em}} = \frac{1}{137}$. (gives $3.65 * 10^5 \mathrm{ms}^{-1}$)

(iv) $r_n = \frac{\varepsilon_0 h^2}{\pi m e^2} n^2$ => $\frac{r_6}{r_1} = \frac{6^2}{1^1} = 36$

$v_n = \frac{nh}{2\pi m r_n}$ => $\frac{v_6}{v_1} = \frac{6}{r_6}\frac{r_1}{1} = \frac{6}{36} = \frac{1}{6}$

[(3 + 2 + 3 + 2) + 4 =14 marks]

Q10.

(a) (i) $t_{\frac{1}{2}} = 3.38 \text{ days} = 2.92 * 10^5 \text{ s}$

$N(t) = N_0 e^{-\mu t}$, $N\left(t_{\frac{1}{2}}\right) = \frac{N_0}{2}$.

⇨ $\frac{N_0}{2} = N_0 e^{-\mu t_{\frac{1}{2}}}$

⇨ $\mu = \frac{\ln(2)}{t_{\frac{1}{2}}}$

$= 0.21 \text{ days}^{-1} = 2.37 * 10^{-6} \text{ s}^{-1}$

(ii) $N(t = 31 \text{ days}) = N_0 e^{-\mu * 31 \text{ days}}$

$= 3 * 10^7 e^{-0.21*31} = 4.47 * 10^4$

(iii) Activity $= \left|\frac{dN}{dt}\right| = \mu N$

at $t = 0$, $\mu N = 0.21 \text{ days}^{-1} * 3 * 10^7 = 6.3 * 10^6 \text{ days}^{-1}$

at $t = 31$ days, $\mu N = 0.21 \text{ days}^{-1} * 4.47 * 10^4 = 9.4 * 10^3 \text{ days}^{-1}$

(iv) For ${}^{222}_{86}Rn$, $A = 222$

$r = (1.2 * 10^{-15}\mathrm{m})\, A^{\frac{1}{3}}$

$= (1.2 * 10^{-15}\mathrm{m})\, 222^{\frac{1}{3}}$

$= 7.27 * 10^{-15}\mathrm{m}$

(b) ${}^{226}_{88}Ra \rightarrow {}^{222}_{86}Rn + {}^{4}_{2}\alpha$

$E_{released} = [m_{Ra} - (m_{Rn} + m_\alpha)]c^2$

$= [226.0254u - 222.0175u - 4.0012u]c^2$

$= 0.0067uc^2$

$1u \simeq m_p$,

⇨ $E_{released} \simeq 0.0067 * 1.67 * 10^{-27}\text{kg} * (3 * 10^{8}\text{ms}^{-1})^{2}$
$\simeq 1.0 * 10^{-12}$ J
$\simeq 6.3 * M\text{eV}$